Bharath Mandha

Machine Learning Engineer | 4+ years building production ML and LLM systems
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EXPERIENCE

Goldman Sachs, AI/ML Engineer **Jun 2025 – Present**

- Designed the retrieval and LangGraph orchestration layers and their FastAPI integration for a RAG platform over 3M+ internal financial documents (10M+ chunks) used weekly by ~73 credit and risk analysts; one of 3 ML engineers on a 7-person team
- Cut average credit-memo preparation from 4.8 to 2.9 hours (~40%), measured on 106 historical and 112 platform-assisted memos, by automating evidence discovery, document analysis and validation with analyst review preserved
- Built a 520-query retrieval benchmark (factual lookups, comparisons, multi-document questions) scored apart from generation and from the 10–20 s end-to-end workflow, then raised Recall@10 from 84.6% to 91.7% and MRR from 0.73 to 0.84 and cut p95 retrieval latency from 1.42 s to 860 ms with 650-token chunks (100 overlap), metadata filtering, FAISS IVF-PQ tuning and a BGE cross-encoder reranking the top 30 for GPT-4o; RAGAS groundedness held above 90%
- Automated 60%+ of targeted repetitive credit-analysis reasoning steps with LangGraph agents; analysts review sensitive decisions
- Designed incremental PySpark and S3 ingestion that keeps the index current with no rebuilds or downtime, tracing each update
- Evaluated quantization and LoRA/QLoRA on Hugging Face models for quality, latency, GPU memory and cost, and showed that infrastructure savings alone didn't justify moving complex credit reasoning to smaller models

Cigna, Machine Learning Engineer **Mar 2022 – Aug 2023**

- Raised 30-day readmission ROC-AUC from 0.846 to 0.912 with XGBoost and LightGBM over 1.5M+ HIPAA-regulated encounters, on a held-out cohort of ~218K encounters (~72K patients) with time-aware, patient-level splits
- At a fixed outreach capacity of 1,000 patients a month, raised high-need patients identified from ~213 to 245 (+15%) through feature engineering, class-imbalance handling and precision-recall threshold optimization
- Engineered 150+ candidate features from labs, demographics, encounter history and clinical notes; ~100 kept in production
- Productionized training, batch inference, registration and retraining (MLflow, Airflow) on SageMaker and EKS through CI/CD with PSI drift monitoring; SHAP explanations let clinical reviewers prioritize thousands of high-risk patients monthly

Best Buy, Associate Data Scientist **Jun 2020 – Feb 2022**

- Cut weighted MAPE of SKU- and store-level demand forecasts from 24.8% to 21.3% (14% relative) with gradient-boosted models benchmarked against time-series baselines on time-based holdouts, tracking RMSE alongside MAPE
- Built PySpark, SQL and BigQuery pipelines over 50M+ transactions across thousands of SKUs and hundreds of stores
- Raised response on comparable campaigns from 4.9% to 5.5% (+12%) with six segments from clustering and cohort analysis

PROJECTS (code, results and CI public on GitHub)

finance-rag-eval, retrieval over SEC filings **2026**

- Page-level retrieval evaluation on FinanceBench (150 questions, 53,399 pages of SEC filings) with a held-out split and bootstrap intervals: a company and period filter (+0.29) and statement-page routing (+0.28) are most of the gain from 0.14 to 0.75 Hit@10
- A MiniLM hybrid led on the dev split but scored 0.73 against BM25's 0.75 on test at 9x the latency, so BM25 ships; the FastAPI service serves ~170 requests/s per core (~$0.07 per million queries), and a browser port matches its scores bit for bit

readmission-capacity, outreach planning on public hospital data **2026**

- Readmission models scored at a fixed call capacity on 101,766 UCI encounters split by patient: a 10% LightGBM call list reaches 24.9% of readmissions (CI 23.2–26.7%) vs 20.5% for prior visits and 15.8% for LACE; calibrated, 547 expected vs 568 observed
- Traced the list's blind spot: 41% of readmissions had no prior visits and it reaches 12% of them. CI re-downloads the data, retrains, and fails if the headline metrics move

SKILLS

Languages and data: Python, SQL, PySpark, BigQuery
ML and LLMs: PyTorch, scikit-learn, XGBoost, LightGBM, SHAP, Hugging Face, LoRA/QLoRA, LangGraph, LangChain, FAISS, RAGAS
MLOps and cloud: MLflow, Airflow, FastAPI, Docker, Kubernetes, Terraform, CI/CD, AWS (SageMaker, S3, EKS), GCP Vertex AI

EDUCATION AND CERTIFICATIONS

University of North Texas, Master of Science in Computer Science **Aug 2023 – May 2025**
AWS Certified Machine Learning Engineer – Associate | Google Cloud Professional Machine Learning Engineer
Databricks Certified Generative AI Engineer Associate